

Module 3

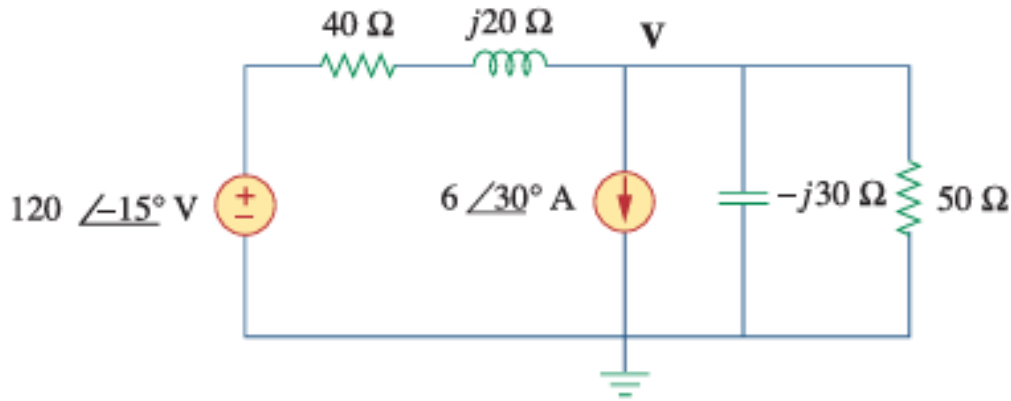
Sinusoidal Steady-State Analysis

Dr. Elizabeth Rufus

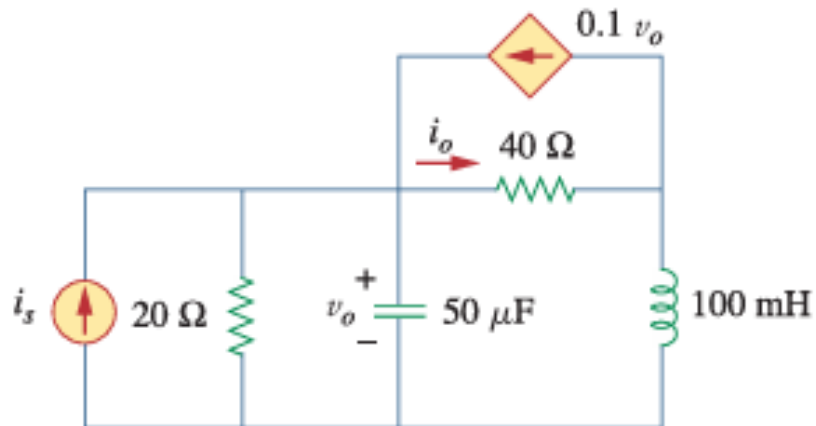
Professor, School of Electronics Engineering

Tutorial

1. Find V in the circuit using Nodal analysis method.

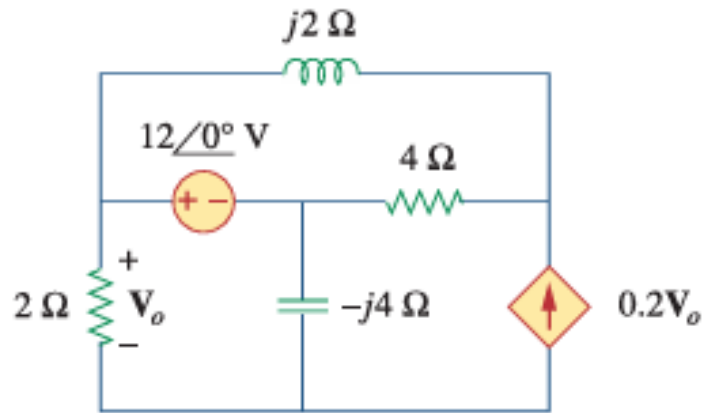


2. Find i_o in the circuit using Nodal analysis method. Let $i_s = 6 \cos(200t + 15^\circ)$

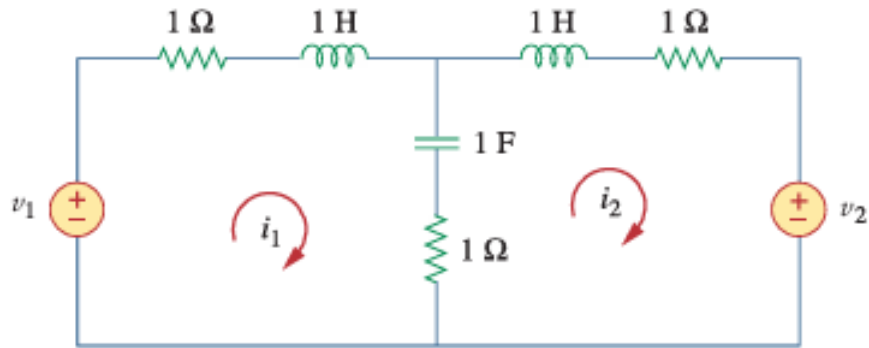


Tutorial

3. Find V_o in the circuit using Nodal analysis method.

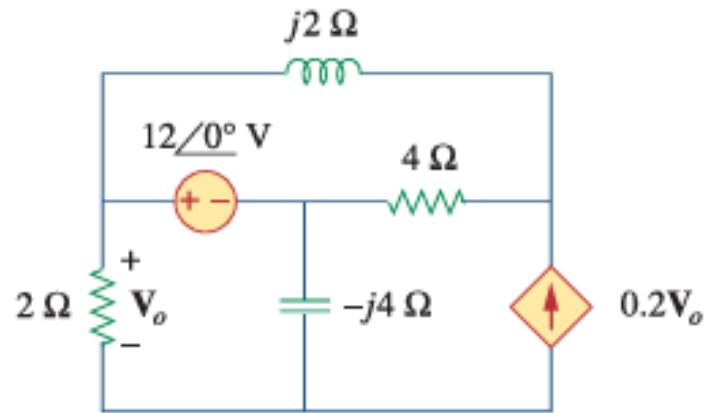


4. In the circuit, determine the mesh currents i_1 and i_2 . Let $v_1 = 10\cos 4t$ V and $v_2 = 20\cos(4t - 30^\circ)$ V.

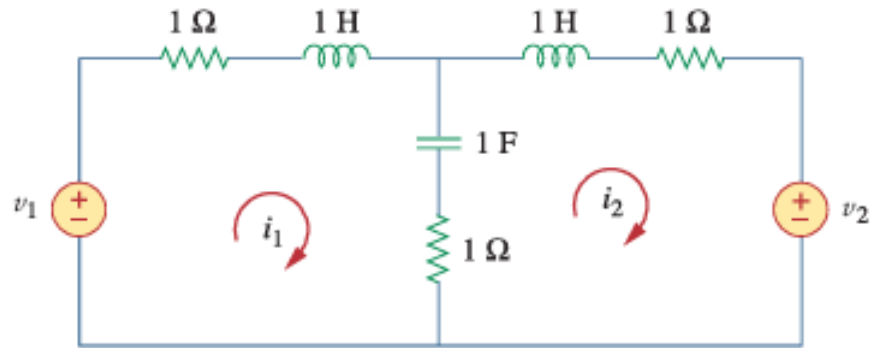


Tutorial

3. Find V_o in the circuit using Nodal analysis method.



4. In the circuit, determine the mesh currents i_1 and i_2 . Let $v_1 = 10\cos 4t$ V and $v_2 = 20\cos(4t - 30^\circ)$ V.



Tutorial

5. Find I_0 in the circuit using Mesh analysis method.

